

Two Output, Integrated VCO, Low-Jitter Clock Generator

FEATURES

- One Single-Ended/Crystal Reference Input Including 24.8832 MHz, 25 MHz, and 26.5625 MHz
- Input Frequency Range: 21.875 MHz to 28.47 MHz
- On-Chip VCO Operates in Frequency Range of 1.75 GHz to 2.05 GHz
- 2x Output Available:
 - Pin-Selectable Between LVPECL, LVDS, or 2-LVCMOS; Operates at 3.3 V
- LVCMOS Bypass Output Available
- Output Frequency Selectable by /1, /2, /3, /4, /6, /8 from a Single Output Divider
- Supports Common LVPECL/LVDS Output Frequencies:
 - 62.5 MHz, 74.25 MHz, 75 MHz, 77.76 MHz, 100 MHz, 106.25 MHz, 125 MHz, 150 MHz, 155.52 MHz, 156.25 MHz, 159.375 MHz, 187.5 MHz, 200 MHz, 212.5 MHz, 250 MHz, 311.04 MHz, 312.5 MHz, 622.08 MHz, 625 MHz
- Supports Common LVCMOS Output Frequencies:
 - 62.5 MHz, 74.25 MHz, 75 MHz, 77.76 MHz, 100 MHz, 106.25 MHz, 125 MHz, 150 MHz, 155.52 MHz, 156.25 MHz, 159.375 MHz, 187.5 MHz, 200 MHz, 212.5 MHz, 250 MHz
- Output Frequency Range: 43.75 MHz to 683.264 MHz (See [Table 3](#))
- Internal PLL Loop Bandwidth: 400 kHz
- High-Performance PLL Core:
 - Phase Noise typically at –146 dBc/Hz at 5-MHz Offset for 625-MHz LVPECL Output
 - Random Jitter typically at 0.509 ps, RMS (10 kHz to 20 MHz) for 625-MHz LVPECL Output
- Output Duty Cycle Corrected to 50% ($\pm 5\%$)
- Low Output Skew of 20 ps on LVPECL Outputs
- Divider Programming Using Control Pins:
 - Two Pins for Prescaler/Feedback Divider
 - Three Pins for Output Divider
 - Two Pins for Output Select

- Chip Enable Control Pin Available
- 3.3-V Core and I/O Power Supply
- Industrial Temperature Range: –40°C to +85°C
- 5-mm × 5-mm, 32-pin, QFN (RHB) Package
- ESD Protection Exceeds 2 kV (HBM)

APPLICATIONS

- Low Jitter Clock Driver for High-End Datacom Applications Including SONET, Ethernet, Fibre Channel, Serial ATA, and HDTV
- Cost-Effective High-Frequency Crystal Oscillator Replacement

DESCRIPTION

The CDCM61002 is a highly versatile, low-jitter frequency synthesizer that can generate two low-jitter clock outputs, selectable between low-voltage positive emitter coupled logic (LVPECL), low-voltage differential signaling (LVDS), or low-voltage complementary metal oxide semiconductor (LVCMOS) outputs, from a low-frequency crystal or LVCMOS input for a variety of wireline and data communication applications. The CDCM61002 features an onboard PLL that can be easily configured solely through control pins. The overall output random jitter performance is less than 1ps, RMS (from 10 kHz to 20 MHz), making this device a perfect choice for use in demanding applications such as SONET, Ethernet, Fibre Channel, and SAN. The CDCM61002 is available in a small, 32-pin, 5-mm × 5-mm QFN package.



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DESCRIPTION, CONTINUED

The CDCM61002 is a high-performance, low phase noise, fully-integrated voltage-controlled oscillator (VCO) clock synthesizer with two universal output buffers that can be configured to be LVPECL, LVDS, or LVC MOS compatible. Each universal output can also be converted to two LVC MOS outputs. Additionally, an LVC MOS bypass output clock is available in an output configuration which can help with crystal loading in order to achieve an exact desired input frequency. It has one fully-integrated, low-noise, LC-based VCO that operates in the 1.75 GHz to 2.05 GHz range.

The phase-locked loop (PLL) synchronizes the VCO with respect to the input, which can either be a low-frequency crystal or a low-noise LVC MOS input. The outputs share an output divider sourced from the VCO core. All device settings are managed through a control pin structure, which has two pins that control the prescaler and feedback divider, three pins that control the output divider, two pins that control the output type, and one pin that controls the output enable. Any time the PLL settings (including the input frequency, prescaler divider, or feedback divider) are altered, a reset must be issued through the Reset control pin (active low for device reset). The reset initiates a PLL recalibration process to ensure PLL lock. When the device is in reset, the outputs and divider are turned off.

The output frequency (f_{OUT}) is proportional to the frequency of the input clock (f_{IN}). The feedback divider, output divider, and VCO frequency set f_{OUT} with respect to f_{IN} . For a configuration setting for common wireline and datacom applications, refer to [Table 2](#). For other applications, use [Equation 1](#) to calculate the exact crystal oscillator frequency required for the desired output.

$$f_{IN} = \left(\frac{\text{Output Divider}}{\text{Feedback Divider}} \right) f_{OUT} \quad (1)$$

The output divider can be chosen from 1, 2, 3, 4, 6, or 8 through the use of control pins. Feedback divider and prescaler divider combinations can be chosen from 25 and 3, 24 and 3, 20 and 4, or 15 and 5, respectively, also through the use of control pins. [Figure 1](#) shows a high-level block diagram of the CDCM61002.

The device operates in a 3.3-V supply environment and is characterized for operation from -40°C to $+85^{\circ}\text{C}$.

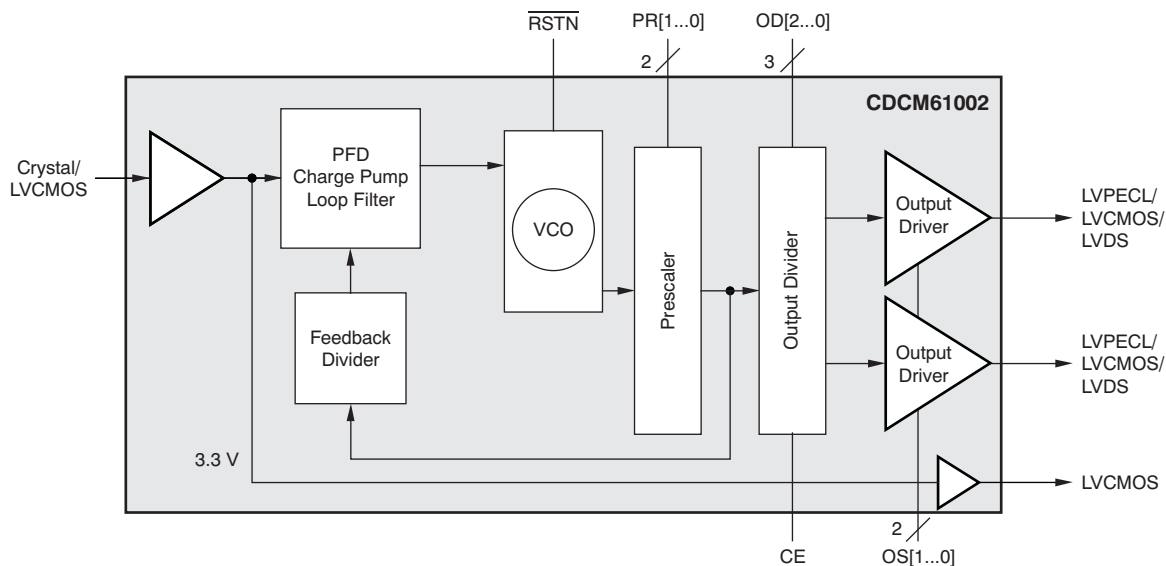


Figure 1. CDCM61002 Block Diagram



This integrated circuit can be damaged by ESD. Texas Instruments recommends that all integrated circuits be handled with appropriate precautions. Failure to observe proper handling and installation procedures can cause damage.

ESD damage can range from subtle performance degradation to complete device failure. Precision integrated circuits may be more susceptible to damage because very small parametric changes could cause the device not to meet its published specifications.

AVAILABLE OPTIONS⁽¹⁾

T _A	PACKAGED DEVICES	FEATURES ⁽²⁾
–40°C to +85°C	CDCM61002RHBT	32-pin QFN (RHB) package, small tape and reel
	CDCM61002RHBR	32-pin QFN (RHB) package, tape and reel

- (1) For the most current specifications and package information, see the Package Option Addendum located at the end of this data sheet or refer to our web site at www.ti.com.
- (2) These packages conform to Lead (Pb)-free and green manufacturing specifications. Additional details including specific material content can be accessed at www.ti.com/leadfree. GREEN: TI defines Green to mean Lead (Pb)-Free and in addition, uses less package materials that do not contain halogens, including bromine (Br), or antimony (Sb) above 0.1% of total product weight. N/A: Not yet available Lead (Pb)-Free; for estimated conversion dates, go to www.ti.com/leadfree. Pb-FREE: TI defines Lead (Pb)-Free to mean RoHS compatible, including a lead concentration that does not exceed 0.1% of total product weight, and, if designed to be soldered, suitable for use in specified lead-free soldering processes.

ABSOLUTE MAXIMUM RATINGS⁽¹⁾

Over operating free-air temperature range (unless otherwise noted).

PARAMETER		VALUE	UNIT
V _{CC_OUT} , V _{CC_PLL1} , V _{CC_PLL2} , V _{CC_VCO} , V _{CC_IN}	Supply voltage range ⁽²⁾	–0.5 to 4.6	V
V _{IN}	Input voltage range ⁽³⁾	–0.5 to (V _{CC_IN} + 0.5)	V
V _{OUT}	Output voltage range ⁽³⁾	–0.5 to (V _{CC_OUT} + 0.5)	V
I _{IN}	Input current	20	mA
I _{OUT}	Output current	50	mA
T _{STG}	Storage temperature range	–65 to +150	°C

- (1) Stresses beyond those listed under *absolute maximum ratings* may cause permanent damage to the device. These are stress ratings only, and functional operation of the device at these or any other conditions beyond those indicated under *recommended operating condition* is not implied. Exposure to absolute-maximum-rated conditions for extended periods may affect device reliability.
- (2) All supply voltages must be supplied simultaneously.
- (3) Input and output negative voltage ratings may be exceeded if the input and output clamp-current ratings are observed.

RECOMMENDED OPERATING CONDITIONS

Over operating free-air temperature range (unless otherwise noted).

PARAMETER		MIN	NOM	MAX	UNIT
V _{CC_OUT}	Output supply voltage	3.0	3.30	3.60	V
V _{CC_PLL1}	PLL supply voltage	3.0	3.30	3.60	V
V _{CC_PLL2}	PLL supply voltage	3.0	3.30	3.60	V
V _{CC_VCO}	On-chip VCO supply voltage	3.0	3.30	3.60	V
V _{CC_IN}	Input supply voltage	3.0	3.30	3.60	V
T _A	Ambient temperature	–40		+85	°C

DISSIPATION RATINGS⁽¹⁾⁽²⁾

PARAMETER		TEST CONDITIONS	VALUE	UNIT
			4 × 4 VIAS ON PAD	
θ _{JA}	Thermal resistance, junction-to-ambient	0 LFM	35	°C/W
θ _{JP} ⁽³⁾	Thermal resistance, junction-to-pad		4	°C/W

(1) The package thermal resistance is calculated in accordance with JESD 51 and JEDEC 2S2P (high-K board).

(2) Connected to GND with nine thermal vias (0.3-mm diameter).

(3) θ_{JP} (junction-to-pad) is used for the QFN package, because the primary heat flow is from the junction to the GND pad of the QFN package.

ELECTRICAL CHARACTERISTICS

At $V_{CC} = 3\text{ V}$ to 3.6 V , $T_A = -40^\circ\text{C}$ to $+85^\circ\text{C}$, unless otherwise noted.

PARAMETER		TEST CONDITIONS	CDCM61002			UNIT
			MIN	TYP	MAX	
LVCMOS Input Characteristics						
f _{IN}	Reference input frequency		21.875		28.47	MHz
V _{IH}	Input high voltage		0.6V _{CC}			V
V _{IL}	Input low voltage				0.4V _{CC}	V
I _{IH}	Input high current	V _{CC} = 3.6 V, V _{IL} = 0 V			200	μA
I _{IL}	Input low current	V _{CC} = 3 V, V _{IH} = 3.6 V			−200	μA
ΔV/ΔT	Reference input edge rate	20% to 80%	0.75			V/ns
DutyREF	Reference input duty cycle		40		60	%
LVCMOS Output Characteristics ⁽¹⁾ (See Figure 9 and Figure 10)						
f _{OSC_OUT}	Bypass output frequency		21.875		28.47	MHz
f _{OUT}	Output frequency		43.75		250	MHz
V _{OH}	Output high voltage	V _{CC} = min to max, I _{OH} = −100 μA	V _{CC} −0.5			V
V _{OL}	Output low voltage	V _{CC} = min to max, I _{OL} = 100 μA		0.3		V
t _{RJIT}	RMS phase jitter	250 MHz (10 kHz to 20 MHz)		0.85		ps, RMS
t _{SLEW-RATE}	Output rise/fall slew rate	20% to 80%	2.4			V/ns
ODC	Output duty cycle		45		55	%
t _{SKEW}	Skew between outputs				50	ps
I _{CC} , LVCMOS	Device current, LVCMOS	f _{IN} = 25 MHz, f _{OUT} = 250 MHz, C _L = 5 pF		120	140	mA
LVPECL Output Characteristics ⁽²⁾ (See Figure 11 and Figure 12)						
f _{OUT}	Output frequency		43.75		683.264	MHz
V _{OH}	Output high voltage		V _{CC} −1.18	V _{CC} −0.73		V
V _{OL}	Output low voltage		V _{CC} −2	V _{CC} −1.55		V
V _{OD}	Differential output voltage		0.6		1.23	V
t _{RJIT}	RMS phase jitter	625 MHz (10 kHz to 20 MHz)		0.77		ps, RMS
t _R /t _F	Output rise/fall time	20% to 80%		175		ps
ODC	Output duty cycle		45		55	%
t _{SKEW}	Skew between outputs				20	ps
I _{CC} , LVPECL	Device current, LVPECL	f _{IN} = 25 MHz, f _{OUT} = 625 MHz		126	144	mA

(1) Figure 9 and Figure 10 show dc and ac test setups, respectively. Jitter measurements made using 25-MHz quartz crystal in.

(2) Figure 11 and Figure 12 show dc and ac test setups, respectively. Jitter measurements made using 25-MHz quartz crystal in.

ELECTRICAL CHARACTERISTICS (continued)

At $V_{CC} = 3\text{ V}$ to 3.6 V , $T_A = -40^\circ\text{C}$ to $+85^\circ\text{C}$, unless otherwise noted.

PARAMETER		TEST CONDITIONS	CDCM61002			UNIT
			MIN	TYP	MAX	
LVDS Output Characteristics ⁽³⁾ (See Figure 13 and Figure 14)						
f _{OUT}	Output frequency		43.75		683.264	MHz
V _{OD}	Differential output voltage		0.247		0.454	V
ΔV _{OD}	V _{DD} magnitude change				50	mV
V _{OS}	Common-mode voltage		1.125		1.375	V
ΔV _{OS}	V _{OS} magnitude change				50	mV
t _{RJIT}	RMS phase jitter	625 MHz (10 kHz to 20 MHz)			0.73	ps, RMS
t _R /t _F	Output rise/fall time	20% to 80%			255	ps
ODC	Output duty cycle		45		55	%
t _{SKEW}	Skew between outputs				30	ps
I _{CC} , LVDS	Device current, LVDS	f _{IN} = 25 MHz, f _{OUT} = 625 MHz		110	125	mA

(3) Figure 13 and Figure 14 show dc and ac test setups, respectively. Jitter measurements made using 25-MHz quartz crystal in.

TYPICAL OUTPUT PHASE NOISE CHARACTERISTICS

Over operating free-air temperature range (unless otherwise noted).

PARAMETER		TEST CONDITIONS	CDCM61002			UNIT
			MIN	TYP	MAX	
250-MHz LVCMOS Output ⁽¹⁾ (see Figure 10)						
phn ₁₀₀	Phase noise at 100-Hz offset			−95		dBc/Hz
phn _{1k}	Phase noise at 1-kHz offset			−110		dBc/Hz
phn _{10k}	Phase noise at 10-kHz offset			−117		dBc/Hz
phn _{100k}	Phase noise at 100-kHz offset			−120		dBc/Hz
phn _{1M}	Phase noise at 1-MHz offset			−135		dBc/Hz
phn _{10M}	Phase noise at 10-MHz offset			−148		dBc/Hz
phn _{20M}	Phase noise at 20-MHz offset			−148		dBc/Hz
t _{RJIT}	RMS phase jitter from 10 kHz to 20 MHz			544		fs, RMS
t _{PJIT}	Total period jitter			27.4		ps, PP
t _{STARTUP}	Start-up time, power supply ramp time of 1 ms, final frequency accuracy of ±10 ppm			2.25		ms
625-MHz LVPECL Output ⁽²⁾ (see Figure 12)						
phn ₁₀₀	Phase noise at 100-Hz offset			−81		dBc/Hz
phn _{1k}	Phase noise at 1-kHz offset			−101		dBc/Hz
phn _{10k}	Phase noise at 10-kHz offset			−109		dBc/Hz
phn _{100k}	Phase noise at 100-kHz offset			−112		dBc/Hz
phn _{1M}	Phase noise at 1-MHz offset			−129		dBc/Hz
phn _{10M}	Phase noise at 10-MHz offset			−146		dBc/Hz
phn _{20M}	Phase noise at 20-MHz offset			−146		dBc/Hz
t _{RJIT}	RMS phase jitter from 10 kHz to 20 MHz			509		fs, RMS
t _{PJIT}	Total period jitter			26.9		ps, PP
t _{STARTUP}	Start-up time, power supply ramp time of 1 ms, final frequency accuracy of ±10 ppm			2.25		ms

(1) Figure 10 shows test setup and uses 25-MHz quartz crystal in, $V_{CC} = 3.3\text{ V}$, $T_A = +25^\circ\text{C}$.

(2) Figure 12 shows test setup and uses 25-MHz quartz crystal in, $V_{CC} = 3.3\text{ V}$, $T_A = +25^\circ\text{C}$.

TYPICAL OUTPUT PHASE NOISE CHARACTERISTICS (continued)

Over operating free-air temperature range (unless otherwise noted).

PARAMETER		TEST CONDITIONS	CDCM61002			UNIT
			MIN	TYP	MAX	
625-MHz LVDS Output ⁽³⁾ (see Figure 14)						
phn ₁₀₀	Phase noise at 100-Hz offset			–88		dBc/Hz
phn _{1k}	Phase noise at 1-kHz offset			–102		dBc/Hz
phn _{10k}	Phase noise at 10-kHz offset			–109		dBc/Hz
phn _{100k}	Phase noise at 100-kHz offset			–112		dBc/Hz
phn _{1M}	Phase noise at 1-MHz offset			–129		dBc/Hz
phn _{10M}	Phase noise at 10-MHz offset			–146		dBc/Hz
phn _{20M}	Phase noise at 20-MHz offset			–146		dBc/Hz
t _{RJIT}	RMS phase jitter from 10 kHz to 20 MHz			510		fs, RMS
t _{PJIT}	Total period jitter			27		ps, PP
t _{STARTUP}	Start-up time, power supply ramp time of 1 ms, final frequency accuracy of ±10 ppm			2.25		ms

(3) Figure 14 shows test setup and uses 25-MHz quartz crystal, V_{CC} = 3.3 V, T_A = +25°C.

TYPICAL OUTPUT JITTER CHARACTERISTICS⁽¹⁾

OUTPUT FREQUENCY (MHz)	INPUT (MHz)	LVCMOS OUTPUT		LVPECL OUTPUT		LVDS OUTPUT	
		t _{RJIT}	t _{PJIT} (pS _{PP})	t _{RJIT}	t _{PJIT} (pS _{PP})	t _{RJIT}	t _{PJIT} (pS _{PP})
62.5	25	592	32.9	611	20.7	667	28.4
75	25	518	27.5	533	19.4	572	25.7
77.76	24.8832	506	29.2	526	20.9	567	26.9
100	25	507	24.5	510	20.7	533	26.5
106.25	26.5625	535	23.5	524	20.2	553	26.5
125	25	557	39.6	556	21.4	570	27.1
150	25	518	38.4	493	18.9	515	26.2
155.52	24.8832	498	36.9	486	19.8	502	26.7
156.25	25	510	37.7	503	20.7	518	26.5
159.375	26.5625	535	37.4	510	19.9	534	26.3
187.5	25	506	32.8	506	20.3	509	25.5
200	25	491	23.3	492	30	499	34.9
212.5	26.5625	520	47.8	509	30.8	530	37.3
250	25	544	27.4	541	21.4	550	27.5
311.04	24.8832			481	20.5	496	24.7
312.5	25			501	20.8	508	25.8
622.08	24.8832			492	27.2	500	27.2
625	25			515	26.9	509	27

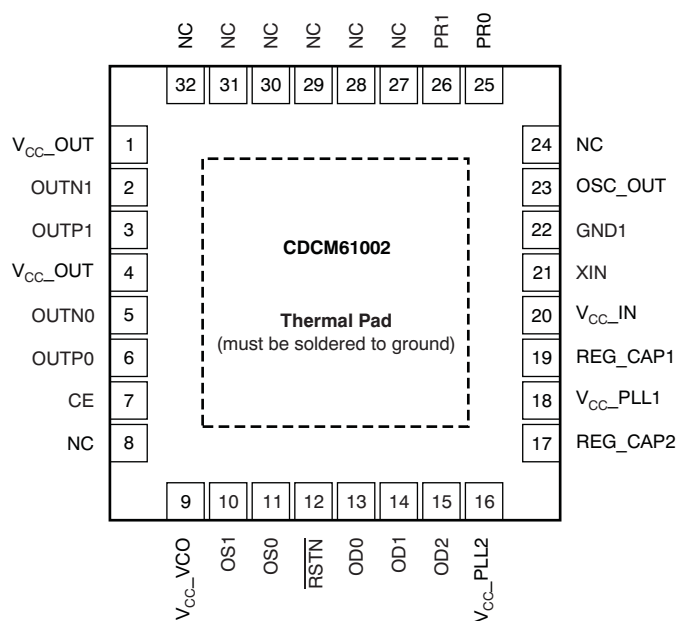
(1) Figure 10, Figure 12, and Figure 14 show LVCMOS, LVPECL, and LVDS test setups (respectively) using appropriate quartz crystal in, V_{CC} = 3.3 V, T_A = +25°C.

CRYSTAL CHARACTERISTICS

PARAMETER	MINIMUM	TYPICAL	MAXIMUM	UNIT
Mode of oscillation	Fundamental			MHz
Frequency	21.875		28.47	MHz
Equivalent series resistance (ESR)			50	Ω
On-chip load capacitance		8	10	pF
Drive level	0.1		1	mW
Maximum shunt capacitance			7	pF

DEVICE INFORMATION

RHB PACKAGE QFN-32 (TOP VIEW)



PIN FUNCTIONS

TERMINAL		TYPE	DIRECTION ⁽¹⁾	DESCRIPTION
NAME	PAD NO.			
VCC_OUT	1, 4	Power		3.3-V supply for the output buffer
VCC_PLL1	18	Power		3.3-V supply for the PLL circuitry
VCC_PLL2	16	Power		3.3-V supply for the PLL circuitry
VCC_VCO	9	Power		3.3-V supply for the internal VCO
VCC_IN	20	Power		3.3-V supply for the input buffers
GND1	22	Ground		Additional ground for device. (GND1 shorted on-chip to GND)
GND	Pad	Ground		Ground is on thermal pad. See Thermal Management .
XIN	21	Input		Parallel resonant crystal/LVCMOS input
OUTP0, OUTN0	6, 5	Output		Differential output pair or two single-ended outputs
OUTP1, OUTN1	3, 2	Output		Differential output pair or two single-ended outputs
OSC_OUT	23	Output		Bypass LVCMOS output
REG_CAP1	19	Output		Capacitor for internal regulator (connect to a 10-μF Y5V capacitor to GND)
REG_CAP2	17	Output		Capacitor for internal regulator (connect to a 10-μF Y5V capacitor to GND)
PR1, PR0	26, 25	Input	Pullup	Prescaler and Feedback divider control pins (see Table 4)
OD2, OD1, OD0	15, 14, 13	Input	Pullup	Output divider control pins (see Table 5)
OS1, OS0	10, 11	Input	Pullup	Output type select control pin (see Table 6)
CE	7	Input	Pullup	Chip enable control pin (see Table 7)
RSTN	12	Input	Pullup	Device reset (active low) (see Table 8)
NC	8, 24, 27, 28, 29, 30, 31, 32			No connection

(1) *Pullup* and *Pulldown* refer to internal input resistors. See [Table 1, Pin Characteristics](#) for typical values.

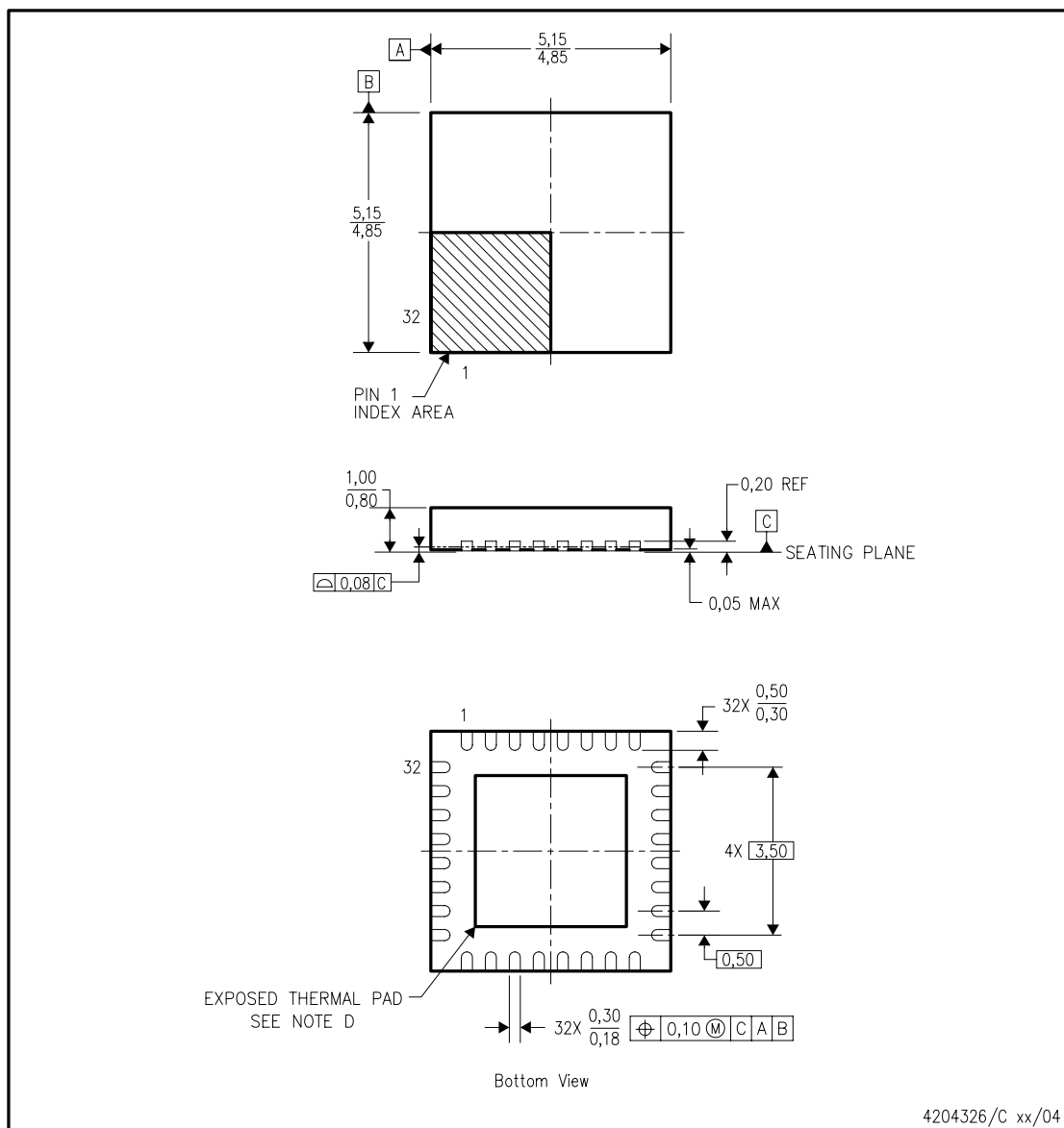
Table 1. PIN CHARACTERISTICS

SYMBOL	PARAMETER	MIN	TYP	MAX	UNIT
C _{IN}	Input capacitance		8	10	pF
R _{PULLUP}	Input pullup resistor		150		kΩ
R _{PULLDOWN}	Input pulldown resistor		150		kΩ

PACKAGE

RHB (S-PQFP-N32)

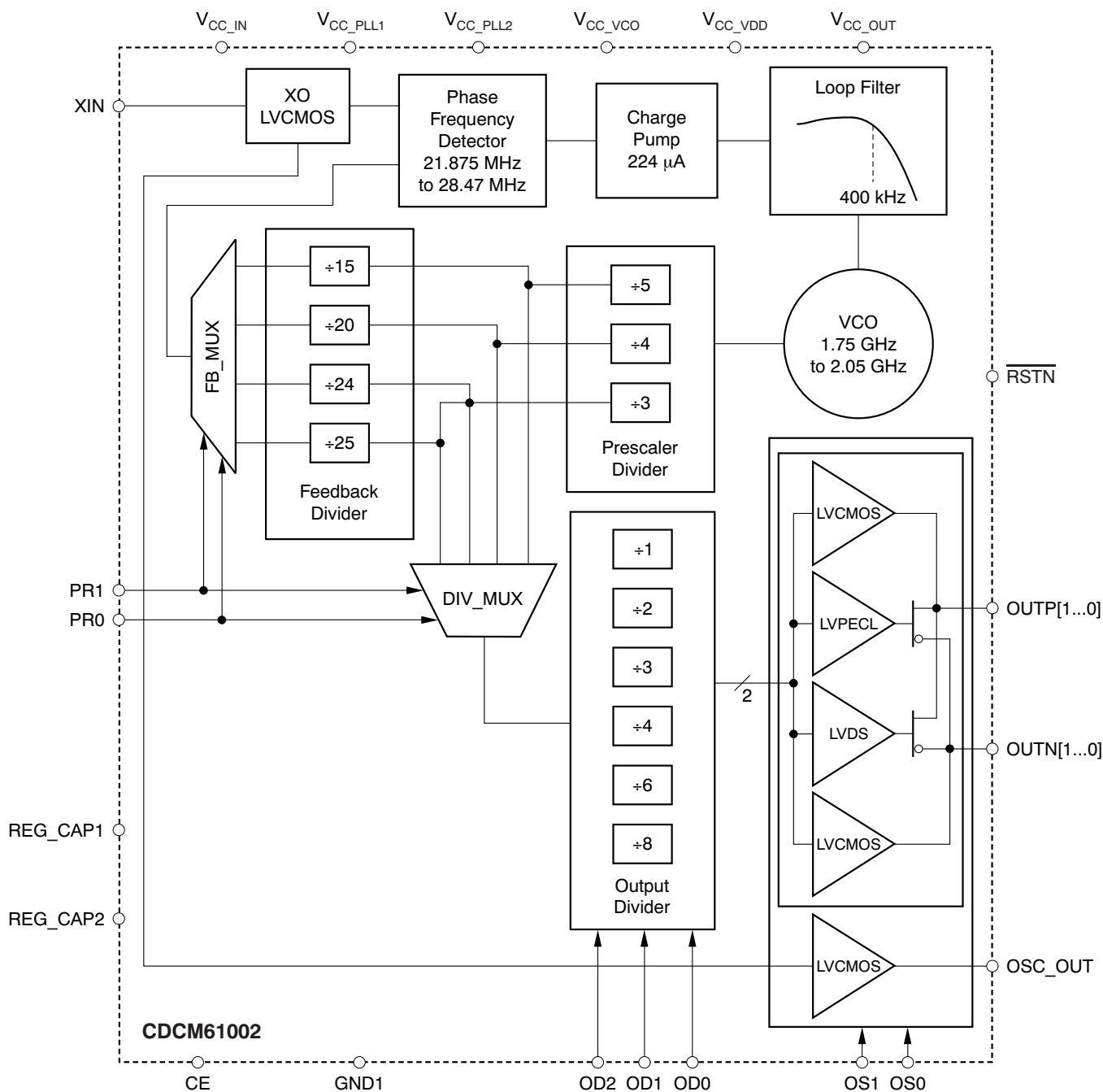
PLASTIC QUAD FLATPACK



- NOTES: A. All linear dimensions are in millimeters.
B. This drawing is subject to change without notice.
C. QFN (Quad Flatpack No-Lead) Package configuration.
D. The Package thermal pad must be soldered to the board for thermal and mechanical performance. See product data sheet for details regarding the exposed thermal pad dimensions.
E. Falls within JEDEC MO-220.

Figure 2. RHB Package

FUNCTIONAL BLOCK DIAGRAM



DEVICE CONFIGURATION

Table 2. Common Configuration

INPUT (MHz)	PRESCALER DIVIDER	FEEDBACK DIVIDER	VCO FREQUENCY (MHz)	OUTPUT DIVIDER	OUTPUT FREQUENCY (MHz)	APPLICATION
25	4	20	2000	8	62.5	GigE
24.75	4	20	1980	8	74.25	HDTV
25	3	24	1800	8	75	SATA
24.8832	3	25	1866.24	8	77.76	SONET
25	3	24	1800	6	100	PCI Express
26.5625	3	24	1912.5	6	106.25	Fibre Channel
25	4	20	2000	4	125	GigE
25	3	24	1800	4	150	SATA
24.8832	3	25	1866.24	4	155.52	SONET
25	3	25	1875	4	156.25	10 GigE
26.5625	3	24	1912.5	4	159.375	10-G Fibre Channel
25	5	15	1875	2	187.5	12 GigE
25	3	24	1800	3	200	PCI Express
26.5625	3	24	1912.5	3	212.5	4-G Fibre Channel
25	4	20	2000	2	250	GigE
24.8832	3	25	1866.24	2	311.04	SONET
25	3	25	1875	2	312.5	XGMII
24.8832	3	25	1866.24	1	622.08	SONET
25	3	25	1875	1	625	10 GigE

Table 3. Generic Configuration

INPUT FREQUENCY RANGE (MHz)	PRESCALER DIVIDER	FEEDBACK DIVIDER	VCO FREQUENCY RANGE (MHz)	OUTPUT DIVIDER	OUTPUT FREQUENCY RANGE (MHz)
21.875 to 25.62	4	20	1750 to 2050	8	54.6875 to 64.05
21.875 to 25.62	4	20	1750 to 2050	6	72.92 to 85.4
21.875 to 25.62	4	20	1750 to 2050	4	109.375 to 128.1
21.875 to 25.62	4	20	1750 to 2050	3	145.84 to 170.8
21.875 to 25.62	4	20	1750 to 2050	2	218.75 to 256.2
21.875 to 25.62	4	20	1750 to 2050	1	437.5 to 512.4
23.33 to 27.33	3	25	1750 to 2050	8	72.906 to 85.408
23.33 to 27.33	3	25	1750 to 2050	6	97.21 to 113.875
23.33 to 27.33	3	25	1750 to 2050	4	145.812 to 170.816
23.33 to 27.33	3	25	1750 to 2050	3	194.42 to 227.75
23.33 to 27.33	3	25	1750 to 2050	2	291.624 to 341.632
23.33 to 27.33	3	25	1750 to 2050	1	583.248 to 683.264
23.33 to 27.33	5	15	1750 to 2050	8	43.75 to 51.25
23.33 to 27.33	5	15	1750 to 2050	6	58.33 to 68.33
23.33 to 27.33	5	15	1750 to 2050	4	87.5 to 102.5
23.33 to 27.33	5	15	1750 to 2050	3	116.66 to 136.66
23.33 to 27.33	5	15	1750 to 2050	2	175 to 205
23.33 to 27.33	5	15	1750 to 2050	1	350 to 410
24.305 to 28.47	3	24	1750 to 2050	8	72.915 to 85.41
24.305 to 28.47	3	24	1750 to 2050	6	97.22 to 113.88

Table 3. Generic Configuration (continued)

INPUT FREQUENCY RANGE (MHz)	PRESCALER DIVIDER	FEEDBACK DIVIDER	VCO FREQUENCY RANGE (MHz)	OUTPUT DIVIDER	OUTPUT FREQUENCY RANGE (MHz)
24.305 to 28.47	3	24	1750 to 2050	4	145.83 to 170.82
24.305 to 28.47	3	24	1750 to 2050	3	194.44 to 227.76
24.305 to 28.47	3	24	1750 to 2050	2	291.66 to 341.64
24.305 to 28.47	3	24	1750 to 2050	1	583.32 to 683.28

Table 4. Programmable Prescaler and Feedback Divider Settings

CONTROL INPUTS		PRESCALER DIVIDER	FEEDBACK DIVIDER	PFD FREQUENCY	
PR1	PR0			MINIMUM	MAXIMUM
0	0	3	24	24.305	28.47
0	1	5	15	23.33	27.33
1	0	3	25	23.33	27.33
1	1	4	20	21.875	25.62

Table 5. Programmable Output Divider

CONTROL INPUTS			OUTPUT DIVIDER
OD2	OD1	OD0	
0	0	0	1
0	0	1	2
0	1	0	3
0	1	1	4
1	0	0	Reserved
1	0	1	6
1	1	0	Reserved
1	1	1	8

Table 6. Programmable Output Type

CONTROL INPUTS		OUTPUT TYPE
OS1	OS0	
0	0	LVC MOS, OSC_OUT Off
0	1	LVDS, OSC_OUT Off
1	0	LVPECL, OSC_OUT Off
1	1	LVPECL, OSC_OUT On

Table 7. Output Enable

CONTROL INPUT	OPERATING CONDITION	OUTPUT
CE		
0	Power Down	Hi-Z
1	Normal	Active

Table 8. Reset

CONTROL INPUT	OPERATING CONDITION	OUTPUT
RSTN		
0	Device Reset	Hi-Z
0 → 1	PLL Recalibration	Hi-Z
1	Normal	Active

TYPICAL CHARACTERISTICS

Over operating free-air temperature range (unless otherwise noted).

TYPICAL CURRENT CONSUMPTION FOR LVPECL OUTPUT vs OUTPUT FREQUENCY

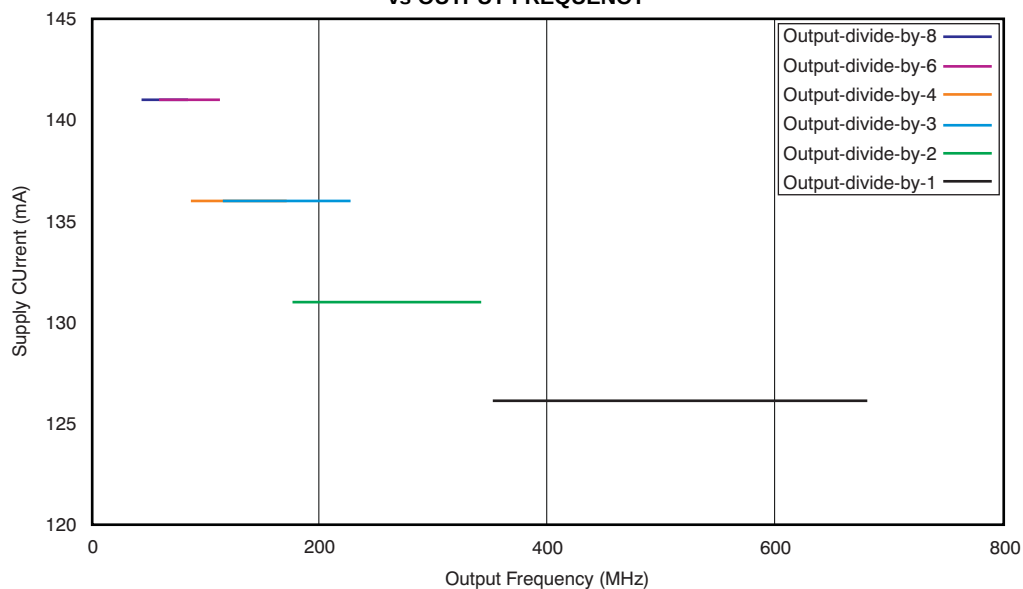


Figure 3.

TYPICAL CURRENT CONSUMPTION FOR LVDS OUTPUT vs OUTPUT FREQUENCY

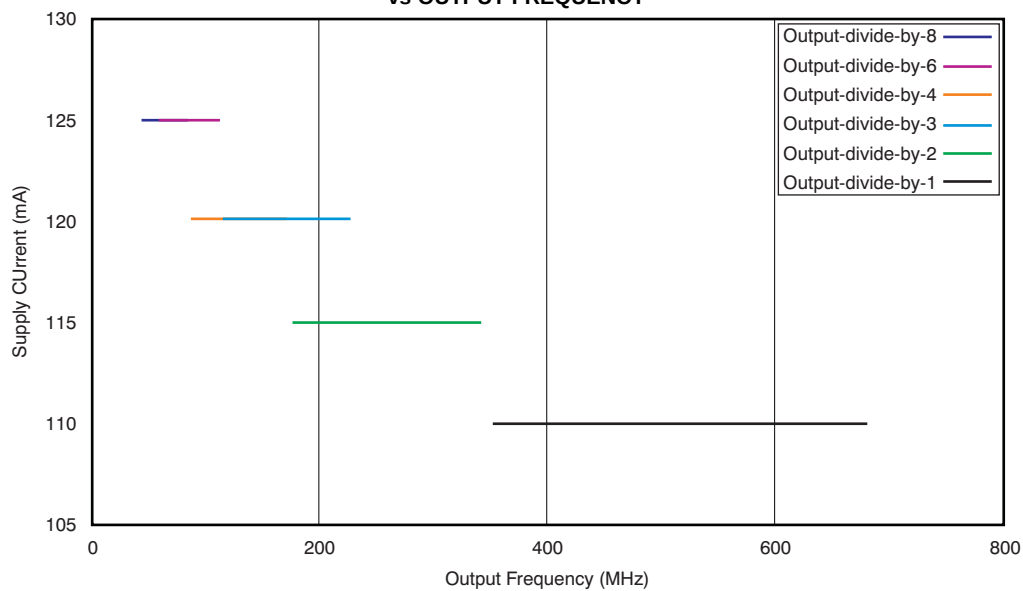


Figure 4.

TYPICAL CHARACTERISTICS (continued)

Over operating free-air temperature range (unless otherwise noted).

**TYPICAL CURRENT CONSUMPTION FOR LVCMOS OUTPUT
WITH 5-pF LOAD vs OUTPUT FREQUENCY**

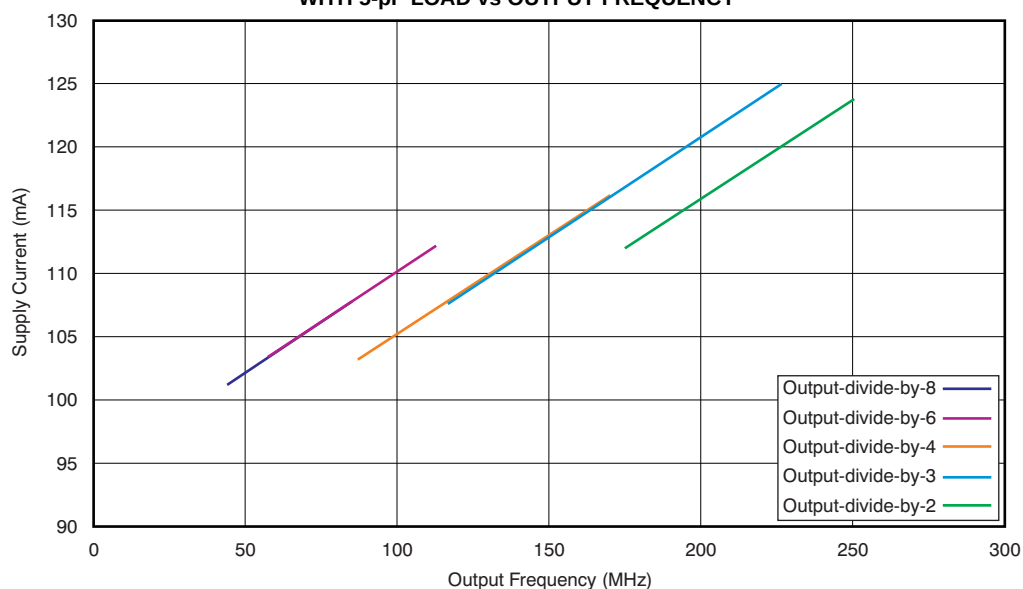


Figure 5.

**TYPICAL LVPECL DIFFERENTIAL OUTPUT VOLTAGE
vs OUTPUT FREQUENCY**

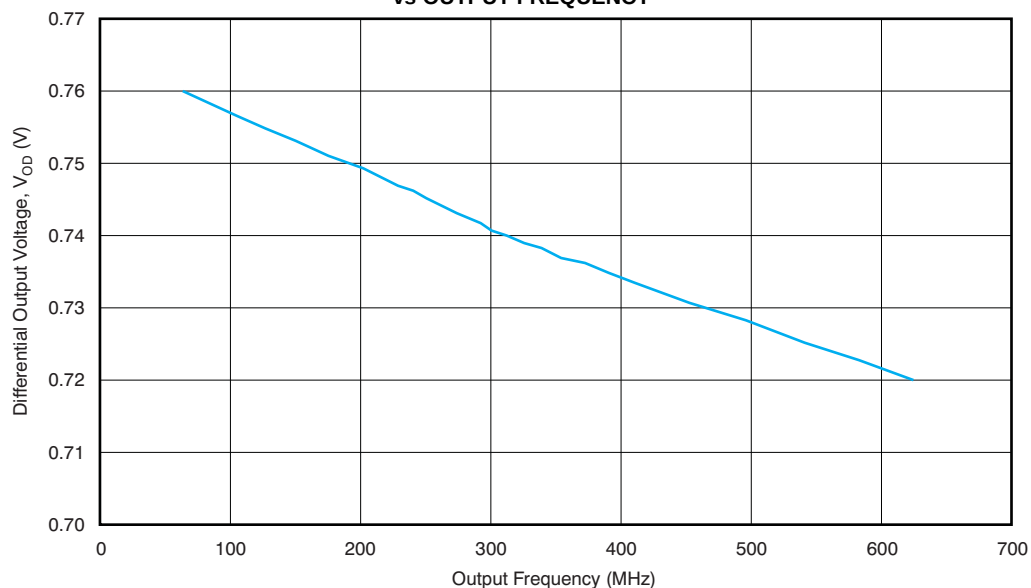


Figure 6.

TYPICAL CHARACTERISTICS (continued)

Over operating free-air temperature range (unless otherwise noted).

**TYPICAL LVDS DIFFERENTIAL OUTPUT VOLTAGE
vs OUTPUT FREQUENCY**

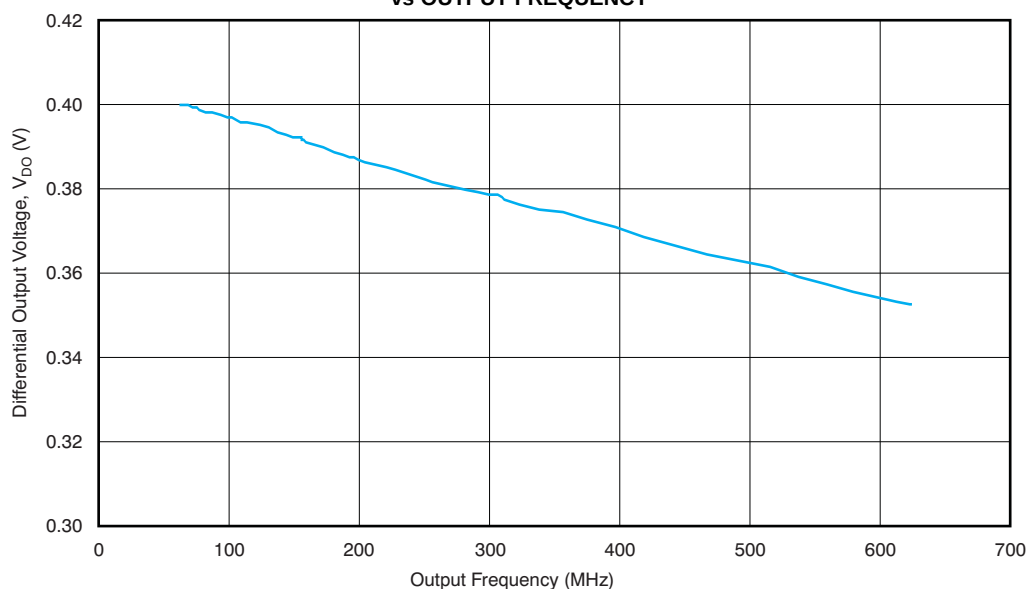


Figure 7.

**TYPICAL LVCMOS OUTPUT VOLTAGE WITH 5-pF LOAD
vs OUTPUT FREQUENCY**

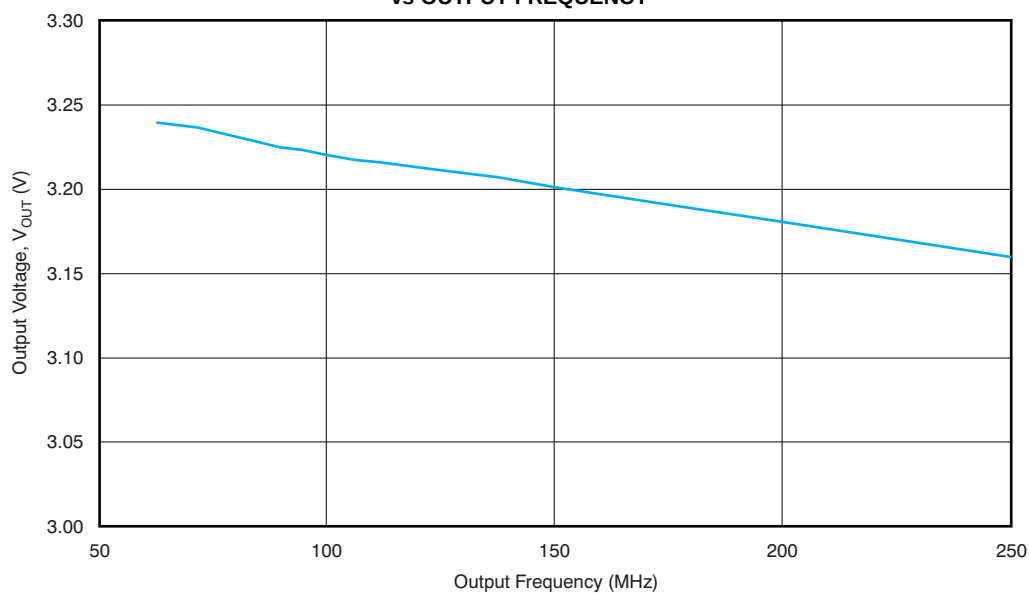


Figure 8.

TEST CONFIGURATIONS

This section describes the function of each block for the CDCM61002. [Figure 9](#) through [Figure 15](#) illustrate how the device should be set up for a variety of output configurations.

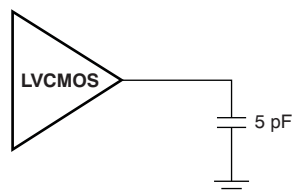


Figure 9. LVC MOS Output Loading During Device Test

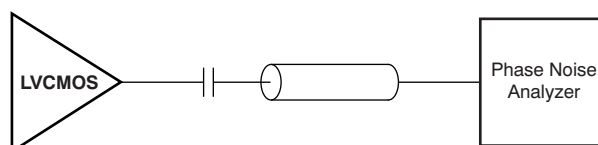


Figure 10. LVC MOS AC Configuration During Device Test

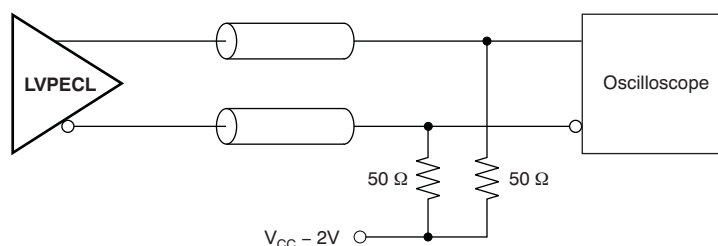


Figure 11. LVPECL DC Configuration During Device Test

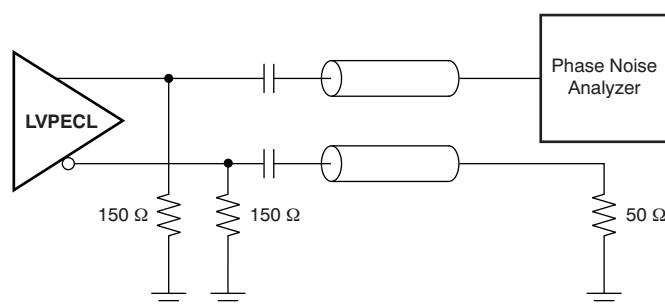


Figure 12. LVPECL AC Configuration During Device Test

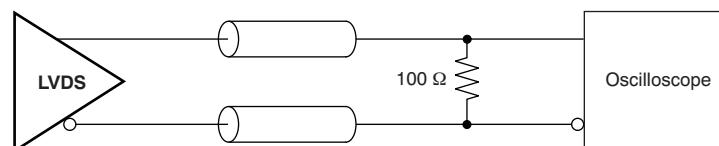


Figure 13. LVDS DC Configuration During Device Test

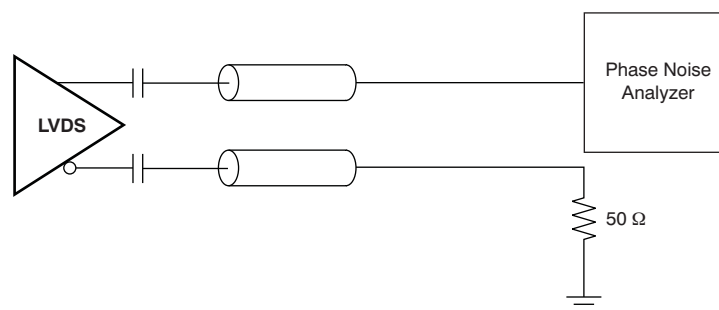


Figure 14. LVDS AC Configuration During Device Test

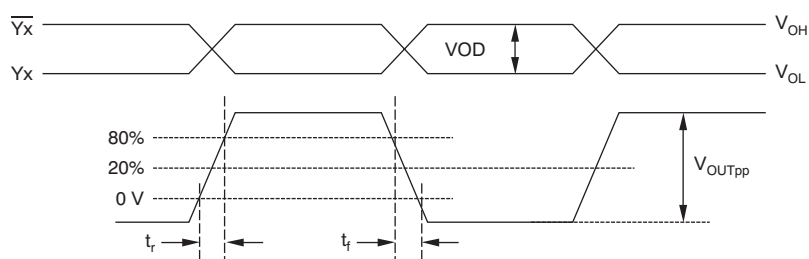


Figure 15. Output Voltage and Rise/Fall Times

FUNCTIONAL DESCRIPTION

Phase-Locked Loop (PLL)

The CDCM61002 includes an on-chip PLL with an on-chip VCO. The PLL blocks consist of a crystal input interface, which can also accept an LVCMOS signal, a phase frequency detector (PFD), a charge pump, an on-chip loop filter, and prescaler and feedback dividers. Completing the CDCM61002 device are the output divider and universal output buffer.

The PLL is powered by on-chip, low-dropout (LDO) linear voltage regulators. The regulated supply network is partitioned such that the sensitive analog supplies are powered from separate LDOs rather than the digital supplies which use a separate LDO regulator. These LDOs provide isolation for the PLL from any noise in the external power-supply rail. The REG_CAP1 and REG_CAP2 pins should each be connected to ground by 10- μ F capacitors to ensure stability.

Configuring the PLL

The CDCM61002 permits PLL configurations to accommodate the various input and output frequencies listed in [Table 2](#) and [Table 3](#). These configurations are accomplished by setting the prescaler divider, feedback divider and output divider. The various dividers are managed by setting the device control pins as shown in [Table 4](#) and [Table 5](#). For each control pin that must be set to a '1', it is recommended to use an external onboard 10-k Ω resistor to the chip supply.

Crystal Input Interface

The recommended oscillation mode for the input crystal is fundamental mode, with a parallel resonance circuit configuration for the crystal.

Crystal load capacitance refers to all capacitances in the oscillator feedback loop. It is equal to the amount of capacitance seen between the terminals of the crystal in the circuit. For parallel resonant mode circuits, the correct load capacitance is necessary to ensure the oscillation of the crystal within the expected parameters.

The CDCM61002 implements an input crystal oscillator circuit architecture that requires the input crystal to interface with one terminal while its other terminal is tied to ground. In this crystal interface, it is important to account for all sources of capacitance when calculating the correct value for the discrete capacitor component, C_L , for a given design.

The CDCM61002 has been characterized with 20-pF parallel resonant crystals. The input crystal oscillator stage in the CDCM61002 is designed to minimize the capacitance required to be added for the 24.8832-MHz, 25-MHz, or 26.5625-MHz crystals to resonate at the correct frequency. The capacitor values shown in [Figure 16](#) were determined using the 20-pF parallel resonant crystals at 24.8832 MHz, 25 MHz, or 26.5625 MHz and checked for minimal ppm error measured at the OSC_OUT pin. This error value can also be calculated from the discrete capacitor component, C_L , required on the XIN pin if the input capacitance on the XIN pin is known (10 pF maximum), in order to match the 20-pF rating. However, it is recommended to use a 10-pF parallel resonant crystal to drive the CDCM61002 and in this case, no additional external capacitors are needed to tune the crystal frequency.

[Table 9](#) lists several recommended crystals and the respective manufacturer of each.

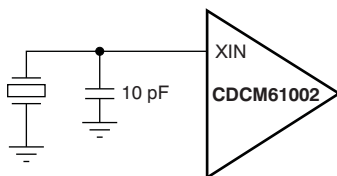


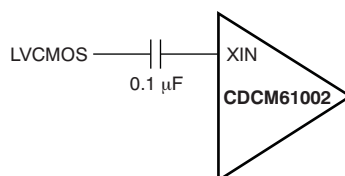
Figure 16. Crystal Input Interface

Table 9. Recommended Crystal Manufacturers

MANUFACTURER	PART NUMBER
Ecliptek	ECCM1B
Epson-Toyocon	TSX-5032
Abrakon	ABM3

LVC MOS Input Interface

Alternately, the CDCM61002 can be operated with an external LVC MOS reference input applied to the XIN pin, which has internal biasing. [Figure 17](#) shows the recommended method to interface an LVC MOS signal with the CDCM61002 through an ac coupling capacitor.

**Figure 17. LVC MOS Input Interface**

Phase Frequency Detector (PFD)

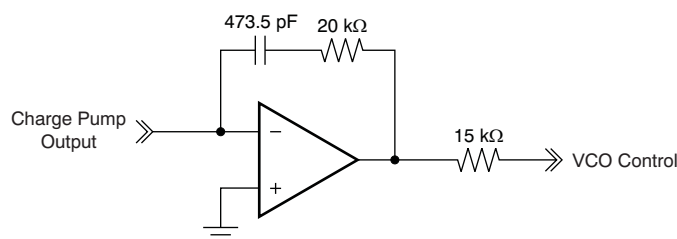
The PFD takes inputs from the input interface and the feedback divider and produces an output that depends on the phase and frequency differences between the two inputs. The allowable range of frequencies at the PFD inputs is 21.875 MHz to 28.47 MHz.

Charge Pump (CP)

The charge pump is controlled by the PFD, which dictates either to pump up or down in order to charge or discharge the integrating section of the on-chip loop filter. The integrated and filtered charge pump current is then converted to a voltage that drives the control voltage node of the internal VCO through the on-chip loop filter. The charge pump current is preset to 224 μ A and cannot be changed.

On-Chip PLL Loop Filter

[Figure 18](#) shows the on-chip active loop filter topology implemented in the device. This design corresponds to a PLL bandwidth of 400 kHz for a PFD in the range of 21.875 MHz to 28.47 MHz, and a charge pump current of 224 μ A.

**Figure 18. On-Chip PLL Loop Filter Topology**

Prescaler Divider and Feedback Divider

The VCO output is routed to the prescaler divider and then to the feedback divider. The prescaler divider and feedback divider are set in tandem with each other, according to the control pin settings given in [Table 4](#). The allowable combinations of the two dividers ensure that the VCO frequency and the PFD frequency are within the specified limits.

On-Chip VCO

The CDCM61002 includes an on-chip, LC oscillator-based VCO with low phase noise covering a frequency range of 1.75 GHz to 2.05 GHz. The VCO must be calibrated to ensure proper operation over the valid device operating conditions. VCO calibration is controlled by a divided-down reference clock input. This calibration requires that the PLL be set up properly to lock the PLL loop and that the reference clock input be present. During the first device initialization after power-up, which occurs after the Power On Reset is released (2.64 V or lower, over valid device operating conditions) or a device reset with the $\overline{\text{RSTN}}$ pin, a VCO calibration sequence is initiated after $16384 \times$ Reference Input Clock Cycles. The VCO calibration then takes about 20 μs over the allowable range of the reference clock input.

The VCO calibration can also be reinitiated with a pulse on the $\overline{\text{RSTN}}$ pin at any time after POR is released on power-up; the RSTN pulse must be at least 100 ns wide.

Output Divider

The output from the prescaler divider is also routed to the output divider. The output divider can be set with control pins according to [Table 5](#).

Output Buffers

Each output buffer can be set to LVPECL or LVDS or 2x LVCMOS, according to [Table 6](#). OSC_OUT is an LVCMOS output that can be used in test mode to monitor proper loading of the input crystal in order to achieve the necessary crystal frequency with the least error. The output buffers are disabled during VCO calibration and are enabled only after calibration is complete.

The output buffers on the CDCM61002 can also be disabled, along with other sections of the device, using the CE pin according to [Table 7](#).

APPLICATION INFORMATION

Start-up Time Estimation

The CDCM61002 startup time can be estimated based on the parameters defined in [Table 10](#) and graphically shown in [Figure 19](#).

Table 10. Start-up Time Dependencies

PARAMETER	DEFINITION	DESCRIPTION	FORMULA/METHOD OF DETERMINATION
t_{REF}	Reference clock period	The reciprocal of the applied reference frequency in seconds.	$t_{REF} = \frac{1}{f_{REF}}$
t_{pul}	Power-up time (low limit)	Power-supply rise time to low limit of Power On Reset (POR) trip point	Time required for power supply to ramp to 2.27 V
t_{puh}	Power-up time (high limit)	Power supply rise time to high limit of POR trip point	Time required for power supply to ramp to 2.64 V
t_{rsu}	Reference start-up time	After POR releases, the Colpits oscillator is enabled. This start-up time is required for the oscillator to generate the requisite signal levels for the delay block to be clocked by the reference input.	500 μ s best-case and 800 μ s worst-case (for a crystal input). 0 s (for an LVCMOS input).
t_{delay}	Delay time	Internal delay time generated from the reference clock. This delay provides time for the reference oscillator to stabilize.	$t_{delay} = 16384 \times t_{ref}$
t_{VCO_CAL}	VCO calibration time	VCO Calibration Time generated from the reference clock. This process selects the operating point for the VCO based on the PLL settings.	$t_{VCO_CAL} = 550 \times t_{ref}$
t_{PLL_LOCK}	PLL lock time	Time required for PLL to lock within ± 10 ppm of f_{REF}	Based on the 400-kHz loop bandwidth, the PLL settles in 5τ or 12.5 μ s.

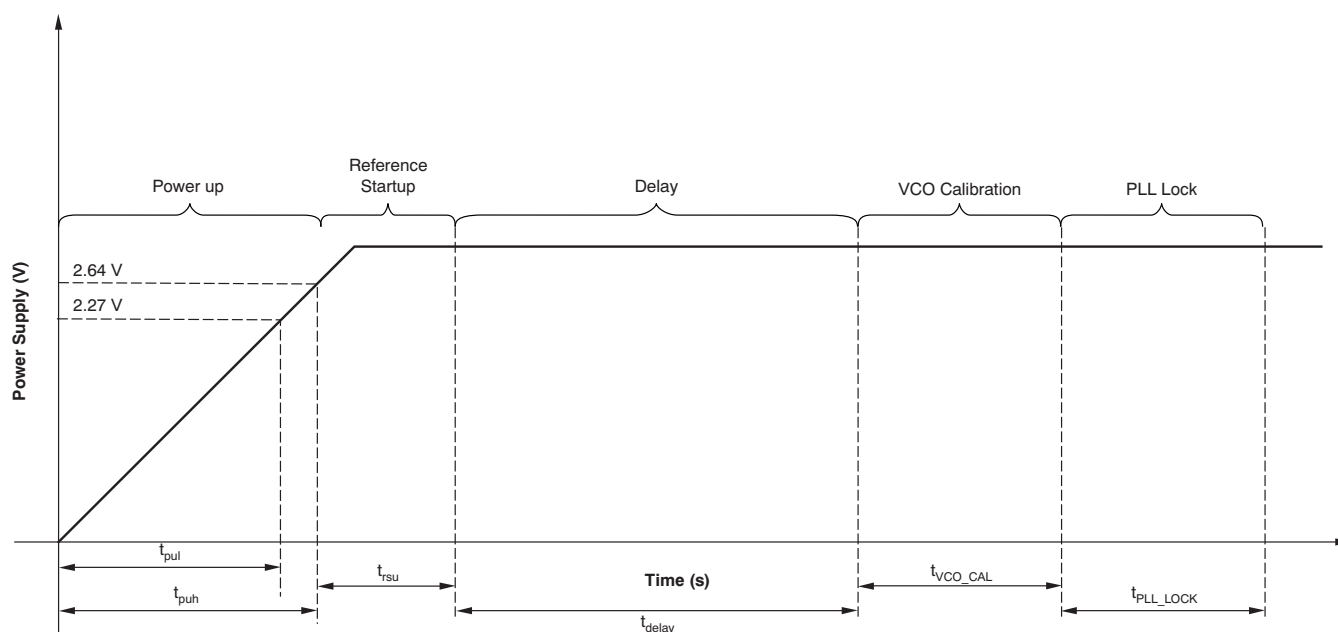


Figure 19. Start-up Time Dependencies

The CDCM61002 start-up time limits, t_{MAX} and t_{MIN} , can be calculated as follows:

$$t_{MAX} = t_{puh} + t_{rsu} + t_{delay} + t_{VCO_CAL} + t_{PLL_LOCK}$$

$$t_{MIN} = t_{pul} + t_{rsu} + t_{delay} + t_{VCO_CAL} + t_{PLL_LOCK}$$

Power Considerations

As a result of the different possible configurations of the CDCM61002, [Table 11](#) is intended to provide enough information on the estimated current consumption of the device. Unless otherwise noted, $V_{CC} = 3.3\text{ V}$ and $T_A = +25^\circ\text{C}$.

Table 11. Estimated Block Power Consumption

BLOCK	CONDITION	CURRENT CONSUMPTION (mA)	IN-DEVICE POWER DISSIPATION (mW)	EXTERNAL RESISTOR POWER DISSIPATION (mW)
Entire device, core current	Output off, no termination resistors	65	214.5	
Output buffer	LVPECL output, active mode	28	42.4	50
	LVC MOS output pair, static	4.5	14.85	
	LVC MOS output pair, transient, 'C _L ' load, 'f' MHz output frequency	$V \times f_{OUT} \times (C_L + 20 \times 10^{-12}) \times 10^3$	$V^2 \times f_{OUT} \times (C_L + 20 \times 10^{-12}) \times 10^3$	
	LVDS output, active mode	20	66	
Divide circuitry	Divide enabled, divide = 1	5	16.5	
	Divide enabled, divide = 2	10	33	
	Divide enabled, divide = 3, 4	15	49.5	
	Divide enabled, divide = 6, 8	20	66	

From [Table 11](#), the current consumption can be calculated for any configuration. For example, the current for the entire device with one LVPECL output in active mode can be calculated by adding up the following blocks: core current, LVPECL output buffer current, and the divide circuitry current. The overall in-device power consumption can also be calculated by summing the in-device power dissipated in each of these blocks.

As an example scenario, let us consider the use case of a crystal input frequency of 25 MHz and device output frequency of 312.5 MHz in LVPECL mode. For this case, the typical overall power dissipation can be calculated as:

$$3.3\text{ V} \times (65 + 2 \times 28 + 10)\text{ mA} = 429\text{ mW}$$

Because the LVPECL output has external resistors and the power dissipated by these resistors is 50 mW, the typical overall in-device power dissipation is:

$$429\text{ mW} - 2 \times 50\text{ mW} = 329\text{ mW}$$

When the LVPECL output is active, the average voltage is approximately 1.9 V on each output as calculated from the LVPECL V_{OH} and V_{OL} specifications. Therefore, the power dissipated in each emitter resistor is approximately $(1.9\text{ V})^2/150\Omega = 25\text{ mW}$.

When the LVC MOS output is active and drives a load capacitance, C_L , the overall LVC MOS output current consumption is the sum of a static pre-driver current and a dynamic switching current (which is a function of the output frequency and the load capacitance).

Let us consider another use case of a crystal input frequency of 26.5625 MHz and device output frequency of 212.5 MHz in LVC MOS mode and driving a 5-pF load capacitance. For this case, the typical overall power dissipation can be calculated as:

$$3.3\text{ V} \times (65 + 15 + 2 \times 21.4)\text{ mA} = 405.24\text{ mW}$$

Thermal Management

Power consumption of the CDCM61002 can be high enough to require attention to thermal management. For reliability and performance reasons, the die temperature should be limited to a maximum of +125°C. That is, as an estimate, T_A (ambient temperature) plus device power consumption times θ_{JA} should not exceed +125°C.

The device package has an exposed pad that provides the primary heat removal path as well as an electrical grounding to the printed circuit board (PCB). To maximize the removal of heat from the package, a thermal land pattern including multiple vias to a ground plane must be incorporated on the PCB within the footprint of the package. The exposed pad must be soldered down to ensure adequate heat conduction out of the package. A recommended land and via pattern is shown in [Figure 20](#).

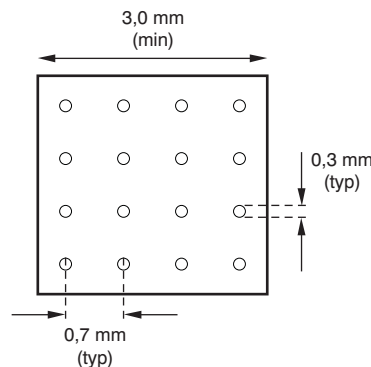


Figure 20. Recommended PCB Layout for CDCM61002

Power-Supply Filtering

PLL-based frequency synthesizers are very sensitive to noise on the power supply, which can dramatically increase the jitter of the PLL. This characteristic is especially true for analog-based PLLs. Thus, it is essential to reduce noise from the system power supply, especially when jitter/phase noise is very critical to applications. A PLL would have attenuated jitter as a result of power-supply noise at frequencies beyond the PLL bandwidth because of attenuation by the loop response.

Filter capacitors are used to eliminate the low-frequency noise from the power supply, where the bypass capacitors provide the very low impedance path for high-frequency noise and guard the power-supply system against the induced fluctuations. These bypass capacitors also provide instantaneous current surges as required by the device and should have low equivalent series resistance (ESR). To properly use these bypass capacitors, they must be placed very close to the power-supply pins and laid out with short loops to minimize inductance. It is recommended to add as many high-frequency (for example, 0.1-μF) bypass capacitors as there are supply pins in the package.

The CDCM61002 power-supply requirements can be grouped into two sets: the analog supply line and the output/input supply line. The analog supply line consists of the following power-supply pins on the CDCM61002: VCC_PLL1, VCC_PLL2, and VCC_VCO. These pins can be shorted together. The output/input supply line consists of the VCC_OUT and the VCC_IN power-supply pins on the CDCM61002. These pins can be shorted together. Inserting a ferrite bead between the analog supply line and the output/input supply line isolates the high-frequency switching noises generated by the device input and outputs, preventing them from leaking into the sensitive analog supply line. Choosing an appropriate ferrite bead with very low dc resistance is important because it is imperative to provide adequate isolation between the sensitive analog supply line and the other board supply lines, and to maintain a voltage at the analog power-supply pins of the CDCM61002 that is greater than the minimum voltage required for proper operation.

Figure 21 shows a general recommendation for decoupling the power supply.

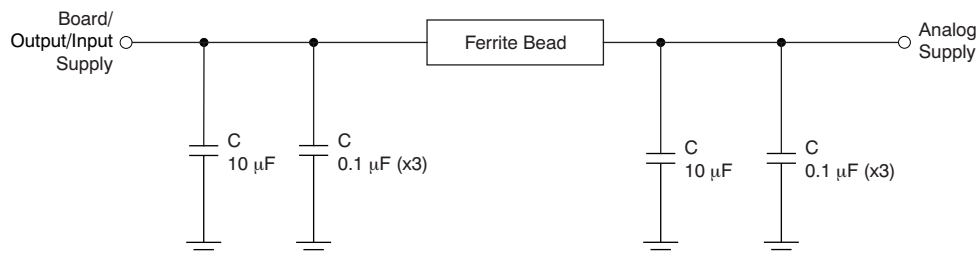


Figure 21. Recommended Power-Supply Decoupling

Output Termination

The CDCM61002 is a 3.3-V clock driver with the following output options: LVPECL, LVDS, or LVCMOS.

LVPECL Termination

The CDCM61002 is an open emitter for LVPECL outputs. Therefore, proper biasing and termination are required to ensure correct operation of the device and to minimize signal integrity. The proper termination for LVPECL is $50\ \Omega$ to $(V_{CC}-2)\text{ V}$, but this dc voltage is not readily available on most PCBs. Thus, a Thevenin equivalent circuit is worked out for the LVPECL termination in both direct-coupled (dc) and ac-coupled cases, as shown in Figure 22 and Figure 23. It is recommended to place all resistive components close to either the driver end or the receiver end.

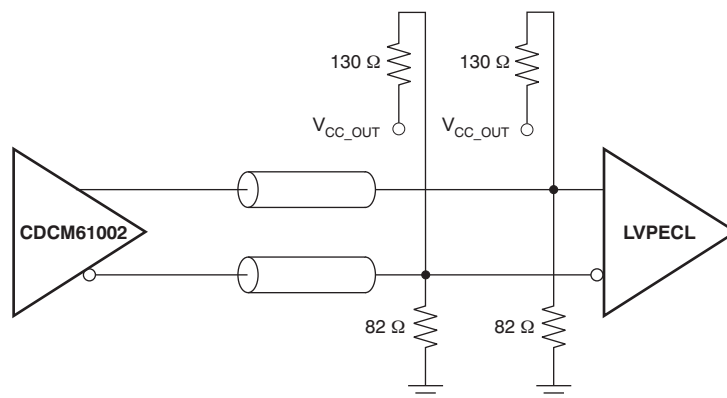


Figure 22. LVPECL Output DC Termination

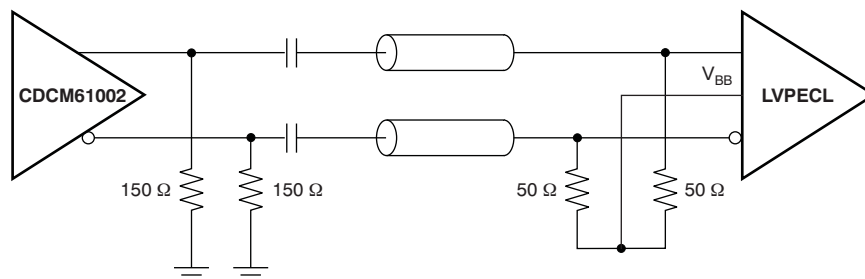


Figure 23. LVPECL Output AC Termination

LVDS Termination

The proper LVDS termination for signal integrity over two 50 Ω lines is 100 Ω between the outputs on the receiver end. Either dc-coupled termination or ac-coupled termination can be used for LVDS outputs, as shown in Figure 24 and Figure 25. It is recommended to place all resistive components close to either the driver end or the receiver end. If the supply voltage of the driver and the receiver are different, ac-coupling is required.

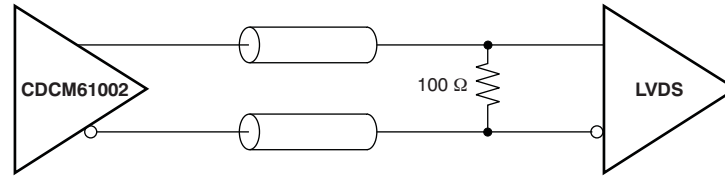


Figure 24. LVDS Output DC Termination

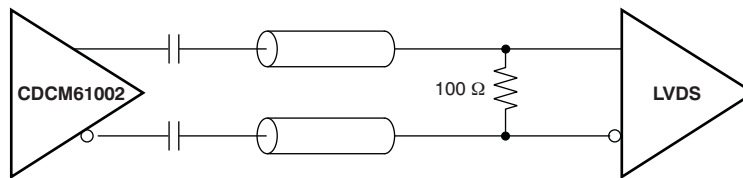


Figure 25. LVDS Output AC Termination

LVC MOS Termination

Series termination is a common technique used to maintain the signal integrity for LVC MOS drivers, if connected to a receiver with a high-impedance input with a pull-up or a pulldown resistor. For series termination, a series resistor (R_S) is placed close to the driver, as shown in Figure 26. The sum of the driver impedance and R_S should be close to the transmission line impedance, which is usually 50 Ω . Because the LVC MOS driver in the CDCM61002 has an impedance of 30 Ω , R_S is recommended to be 22 Ω to maintain proper signal integrity.

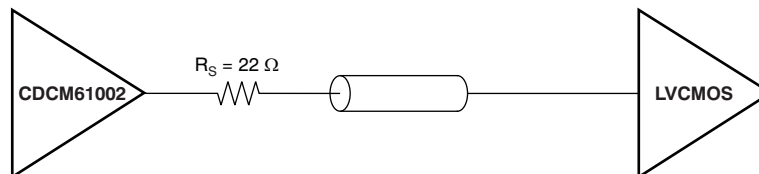


Figure 26. LVC MOS Output Termination

Interfacing Between LVPECL and HCSL

Because the LVPECL common-mode voltage is different from the HCSL common-mode voltage, ac-coupled termination is used. The 150-Ω resistor ensures proper biasing of the CDCM61002 LVPECL output stage, while the 471-Ω and 56-Ω resistor network biases the HCSL receiver input stage, as shown in Figure 27.

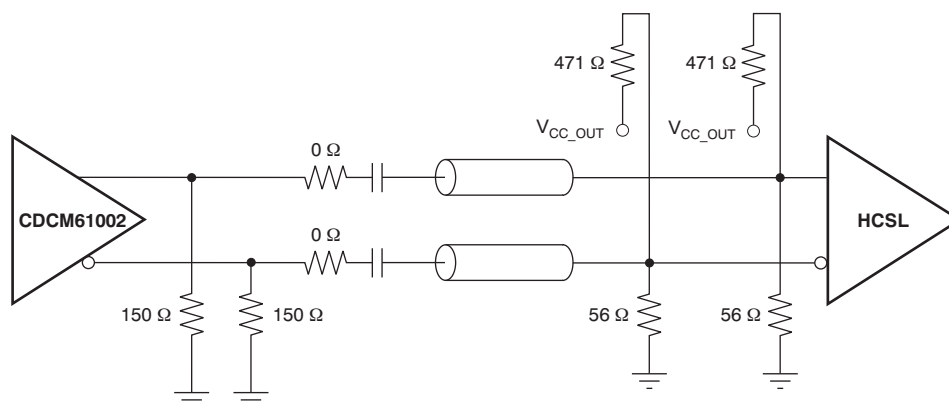


Figure 27. LVPECL to HCSL Interface

Revision History

NOTE: Page numbers for previous revisions may differ from page numbers in the current version.

Changes from Revision A (May, 2009) to Revision B	Page
• Added sentence about parallel resonant crystal to last paragraph of <i>Crystal Input Interface</i> section.....	19
• Updated Figure 22	25
• Revised text in <i>LVC MOS Termination</i> section.....	26
• Updated Figure 27	27
Changes from Original (February, 2009) to Revision A	Page
• Added <i>Interfacing Between LVPECL and HCSL</i> section.....	27

PACKAGING INFORMATION

Orderable Device	Status ⁽¹⁾	Package Type	Package Drawing	Pins	Package Qty	Eco Plan ⁽²⁾	Lead/Ball Finish	MSL Peak Temp ⁽³⁾
CDCM61002RHBR	ACTIVE	QFN	RHB	32	3000	Green (RoHS & no Sb/Br)	CU NIPDAU	Level-2-260C-1 YEAR
CDCM61002RHBT	ACTIVE	QFN	RHB	32	250	Green (RoHS & no Sb/Br)	CU NIPDAU	Level-2-260C-1 YEAR

⁽¹⁾ The marketing status values are defined as follows:

ACTIVE: Product device recommended for new designs.

LIFEBUY: TI has announced that the device will be discontinued, and a lifetime-buy period is in effect.

NRND: Not recommended for new designs. Device is in production to support existing customers, but TI does not recommend using this part in a new design.

PREVIEW: Device has been announced but is not in production. Samples may or may not be available.

OBsolete: TI has discontinued the production of the device.

⁽²⁾ Eco Plan - The planned eco-friendly classification: Pb-Free (RoHS), Pb-Free (RoHS Exempt), or Green (RoHS & no Sb/Br) - please check <http://www.ti.com/productcontent> for the latest availability information and additional product content details.

TBD: The Pb-Free/Green conversion plan has not been defined.

Pb-Free (RoHS): TI's terms "Lead-Free" or "Pb-Free" mean semiconductor products that are compatible with the current RoHS requirements for all 6 substances, including the requirement that lead not exceed 0.1% by weight in homogeneous materials. Where designed to be soldered at high temperatures, TI Pb-Free products are suitable for use in specified lead-free processes.

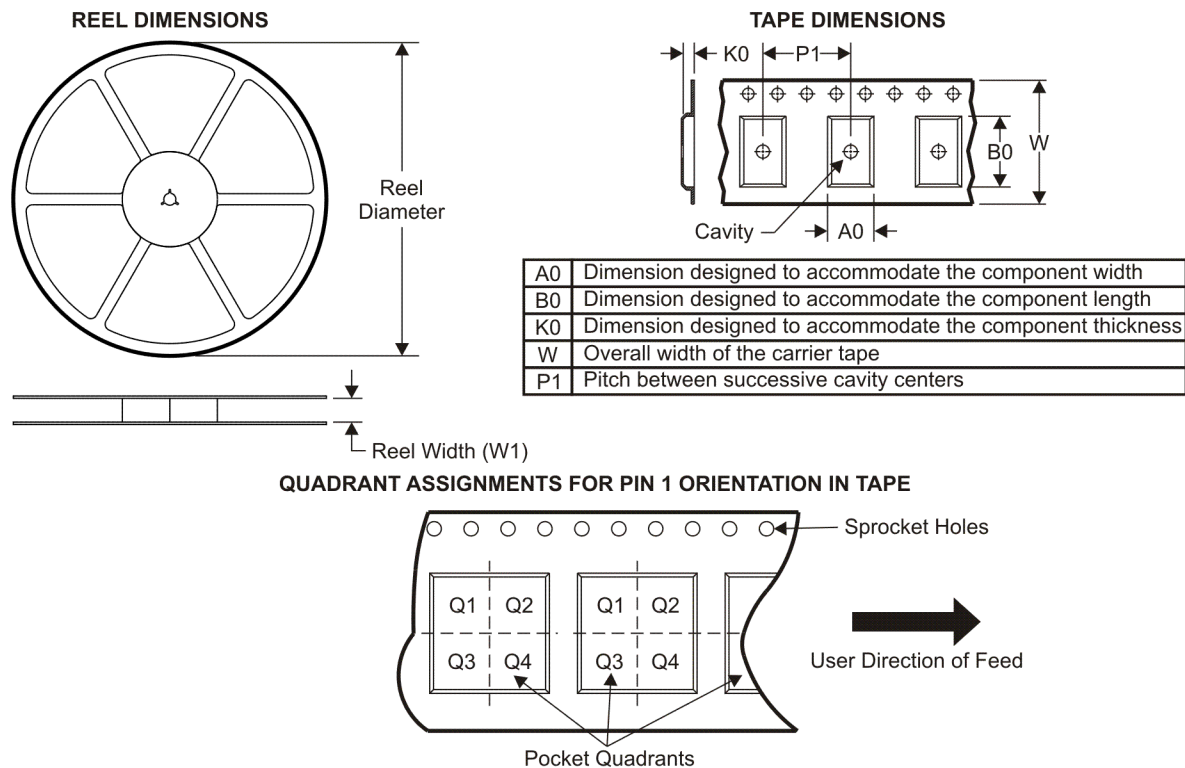
Pb-Free (RoHS Exempt): This component has a RoHS exemption for either 1) lead-based flip-chip solder bumps used between the die and package, or 2) lead-based die adhesive used between the die and leadframe. The component is otherwise considered Pb-Free (RoHS compatible) as defined above.

Green (RoHS & no Sb/Br): TI defines "Green" to mean Pb-Free (RoHS compatible), and free of Bromine (Br) and Antimony (Sb) based flame retardants (Br or Sb do not exceed 0.1% by weight in homogeneous material)

⁽³⁾ MSL, Peak Temp. -- The Moisture Sensitivity Level rating according to the JEDEC industry standard classifications, and peak solder temperature.

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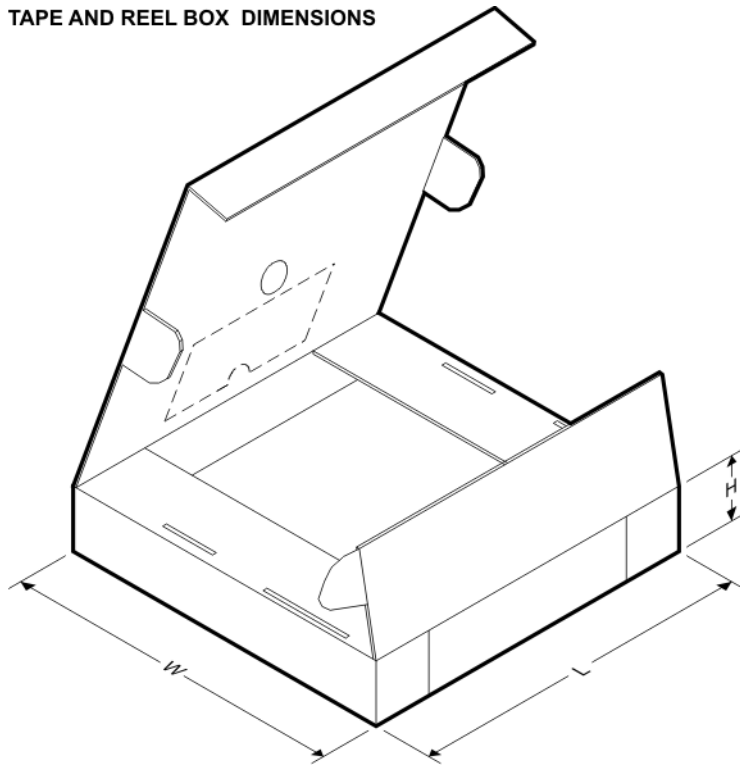
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TAPE AND REEL INFORMATION


*All dimensions are nominal

Device	Package Type	Package Drawing	Pins	SPQ	Reel Diameter (mm)	Reel Width W1 (mm)	A0 (mm)	B0 (mm)	K0 (mm)	P1 (mm)	W (mm)	Pin1 Quadrant
CDCM61002RHBR	QFN	RHB	32	3000	330.0	12.4	5.3	5.3	1.5	8.0	12.0	Q2
CDCM61002RHBT	QFN	RHB	32	250	180.0	12.4	5.3	5.3	1.5	8.0	12.0	Q2

TAPE AND REEL BOX DIMENSIONS

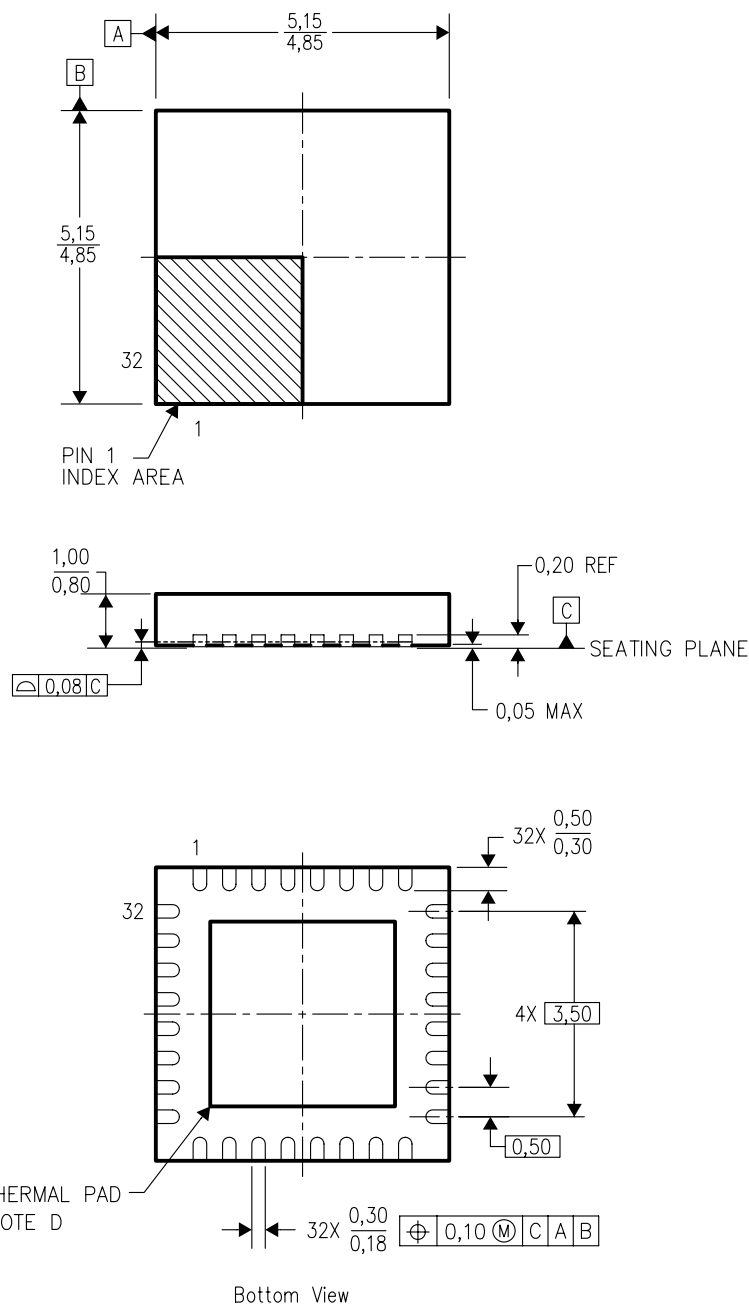


*All dimensions are nominal

Device	Package Type	Package Drawing	Pins	SPQ	Length (mm)	Width (mm)	Height (mm)
CDCM61002RHBR	QFN	RHB	32	3000	346.0	346.0	29.0
CDCM61002RHBT	QFN	RHB	32	250	190.5	212.7	31.8

RHB (S-PQFP-N32)

PLASTIC QUAD FLATPACK



4204326/C xx/04

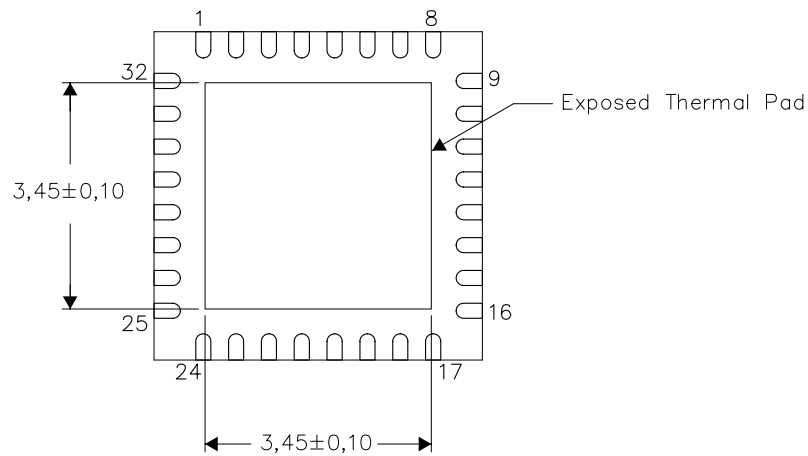
- NOTES:
- All linear dimensions are in millimeters.
 - This drawing is subject to change without notice.
 - QFN (Quad Flatpack No-Lead) Package configuration.
 - The Package thermal pad must be soldered to the board for thermal and mechanical performance. See product data sheet for details regarding the exposed thermal pad dimensions.
 - Falls within JEDEC MO-220.

THERMAL INFORMATION

This package incorporates an exposed thermal pad that is designed to be attached directly to an external heatsink. The thermal pad must be soldered directly to the printed circuit board (PCB). After soldering, the PCB can be used as a heatsink. In addition, through the use of thermal vias, the thermal pad can be attached directly to the appropriate copper plane shown in the electrical schematic for the device, or alternatively, can be attached to a special heatsink structure designed into the PCB. This design optimizes the heat transfer from the integrated circuit (IC).

For information on the Quad Flatpack No-Lead (QFN) package and its advantages, refer to Application Report, QFN/SON PCB Attachment, Texas Instruments Literature No. SLUA271. This document is available at www.ti.com.

The exposed thermal pad dimensions for this package are shown in the following illustration.

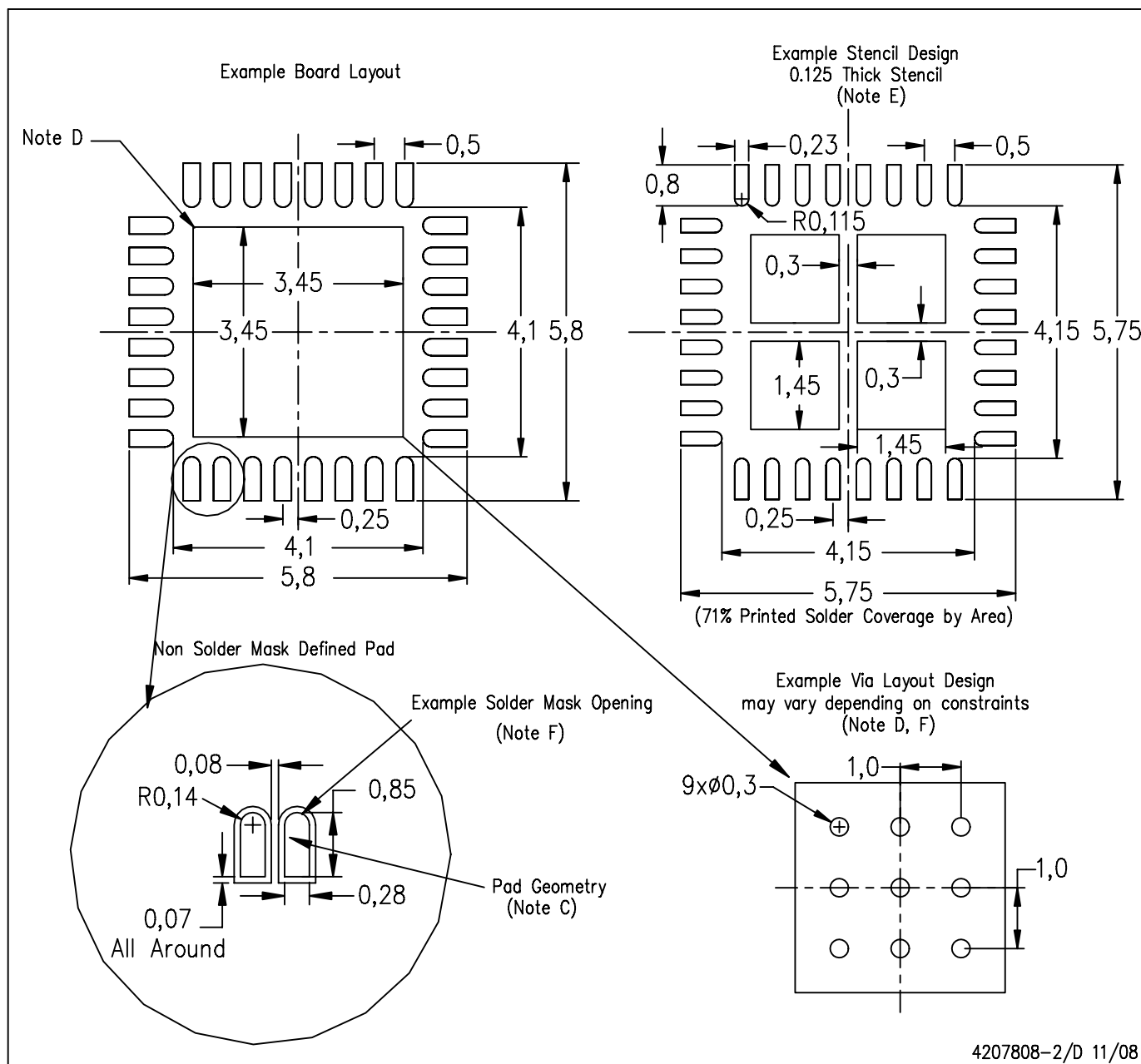


Bottom View

NOTE: All linear dimensions are in millimeters

Exposed Thermal Pad Dimensions

RHB (S-PQFP-N32)



4207808-2/D 11/08

- NOTES:
- All linear dimensions are in millimeters.
 - This drawing is subject to change without notice.
 - Publication IPC-7351 is recommended for alternate designs.
 - This package is designed to be soldered to a thermal pad on the board. Refer to Application Note, Quad Flat-Pack Packages, Texas Instruments Literature No. SCBA017, SLUA271, and also the Product Data Sheets for specific thermal information, via requirements, and recommended board layout. These documents are available at www.ti.com <<http://www.ti.com>>.
 - Laser cutting apertures with trapezoidal walls and also rounding corners will offer better paste release. Customers should contact their board assembly site for stencil design recommendations. Refer to IPC 7525 for stencil design considerations.
 - Customers should contact their board fabrication site for recommended solder mask tolerances and via tenting recommendations for vias placed in the thermal pad.

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